



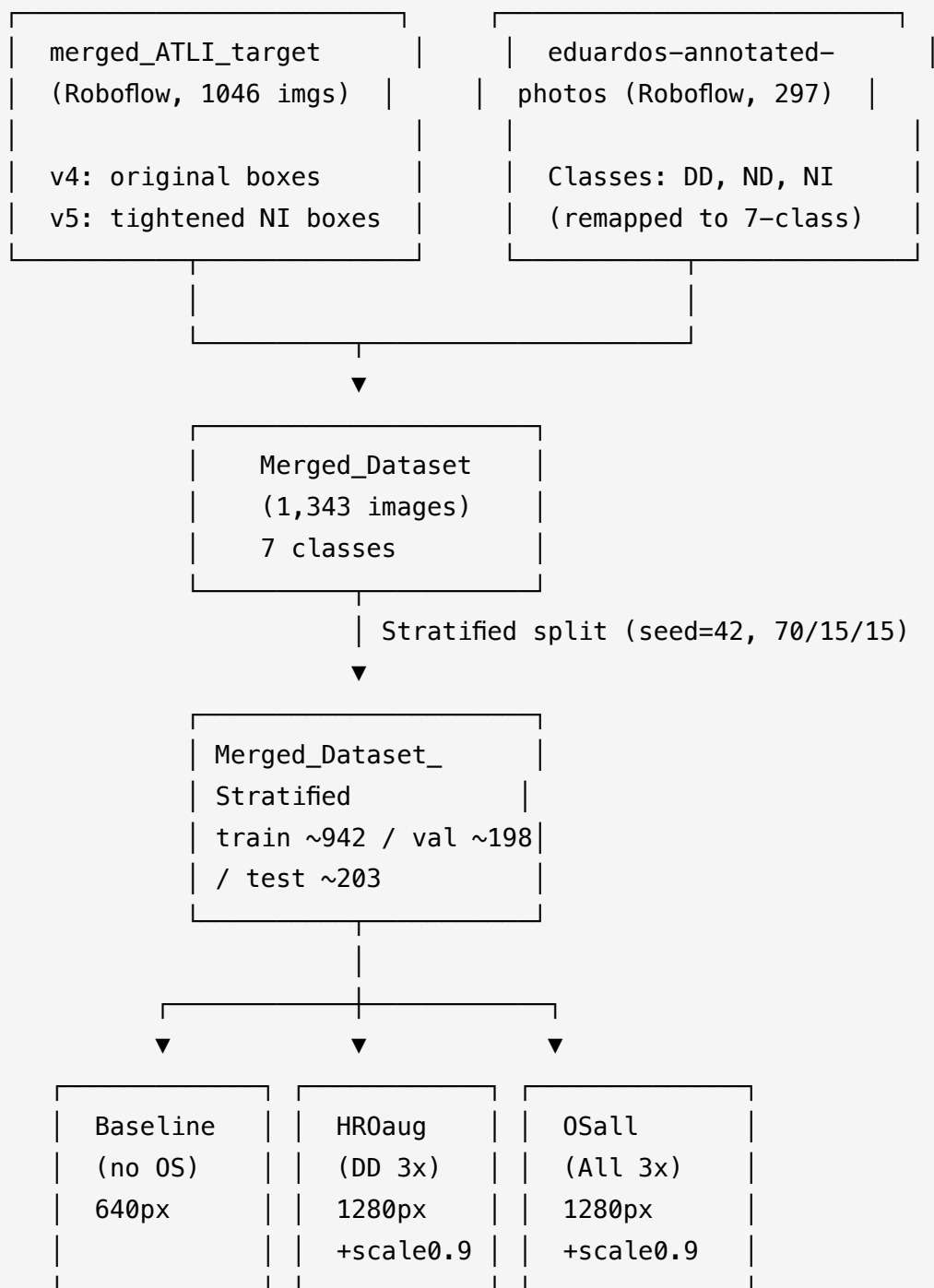
Experiment Writeup

Date: June 24, 2026

Summary: Benchmarked YOLOv5n, YOLOv8n, and YOLOv11n across three training recipes using 5-fold cross-validation. Best result: **mAP 80.6%** (YOLOv11n with OSall), up from the paper's 78.9%. This is a CV mean, not a single-split number.

Data Pipeline Overview

Dataset Construction



Oversampling Strategy

Train images containing class:

Baseline (no change)	Broken_Insulator (139 imgs)	┌	x3 copies (2 extra)	
	Defective_Damper (177 imgs)			
	Flashover_Insulator (302 imgs)			
	Self-Exploded_Ins (385 imgs)	└		
	Normal_Damper (1586 imgs)			not oversampled
	Normal_Insulators (1275 imgs)			not oversampled
	Birdnest (167 imgs)			not oversampled

HR0aug: only Defective_Damper images x3 -> DD 177 -> 531 effective
OSall: all 4 defect classes x3 -> ~1,003 imgs -> ~2,539 effective

Annotation Versions

merged_ATLI_target on Roboflow:

v4 (original)	----->	v5 (tightened)
NI: 873 boxes		NI: 886 boxes (+13)
ND: 1460 boxes		ND: 1461 boxes (+1)
all others same		all others same
Loose NI boxes (some overlap with defects)		Tight NI polygons (cleaner boundaries, better normal/defect discrimination)

Used in:

- All CV experiments
- All single-split seeds
- Paper comparison

Used in:

- A/B comparison

Figures

All figures in `results/figures/writeup/` .

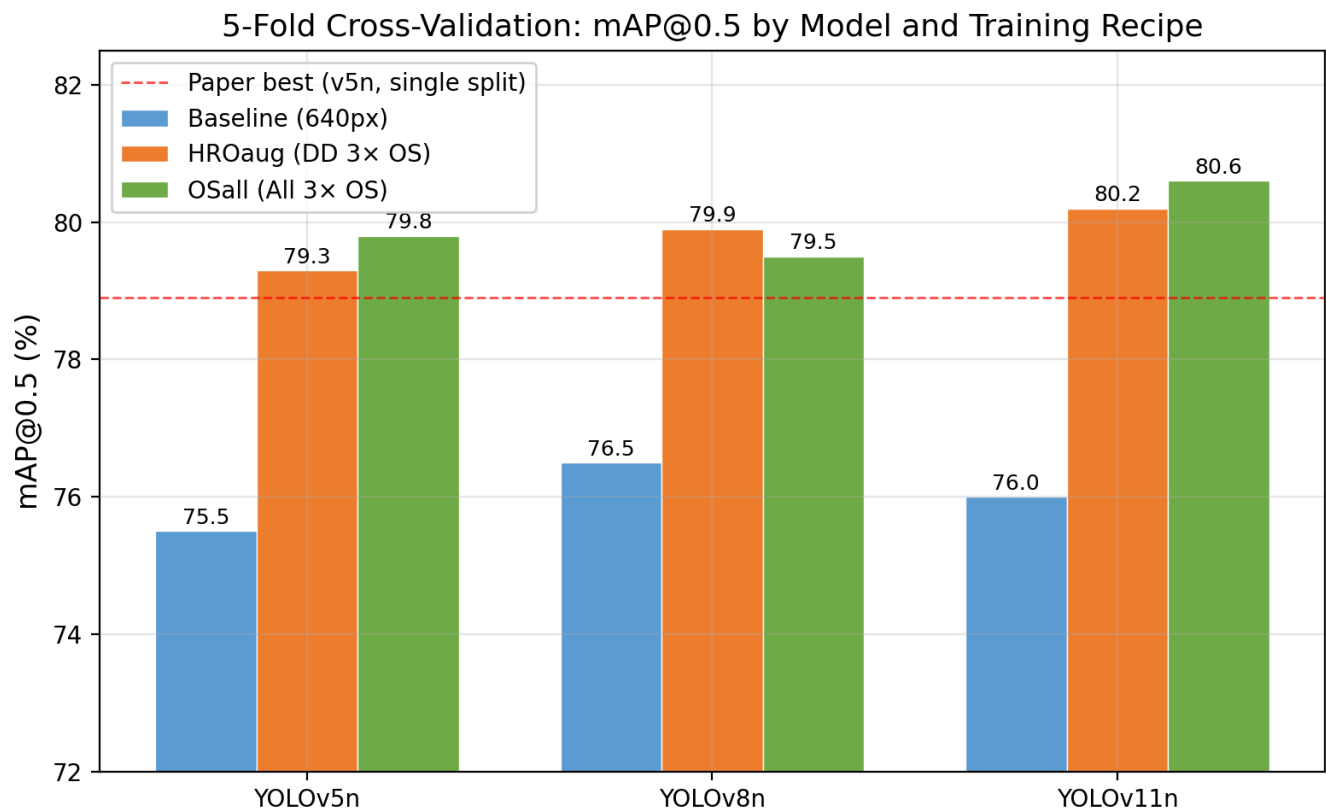


Fig 1: 5-fold CV mAP@0.5 across all models and recipes. Red dashed line = paper's best (78.9%, single split).

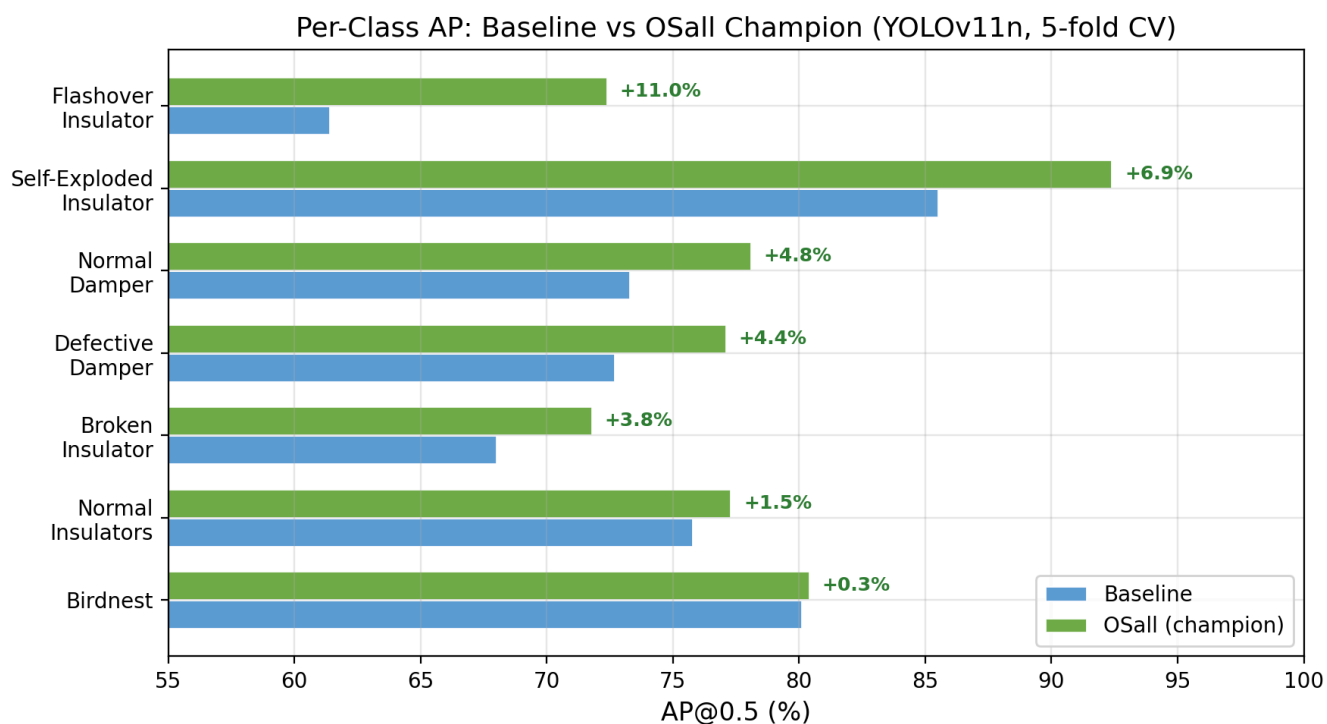


Fig 2: Per-class AP improvement from baseline to OSall (YOLOv11n). Flashover had the largest gain (+11.0%).

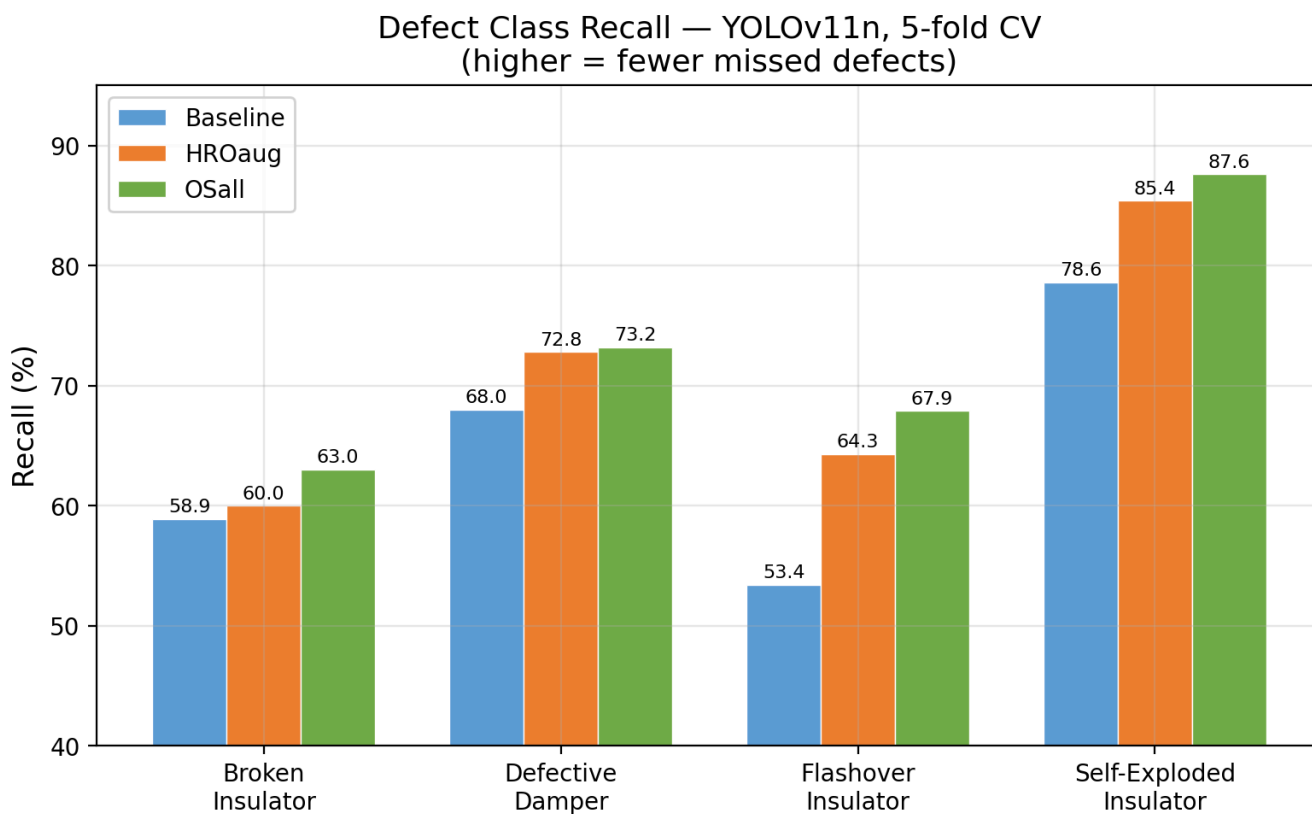


Fig 3: Defect class recall. Higher = fewer missed defects.

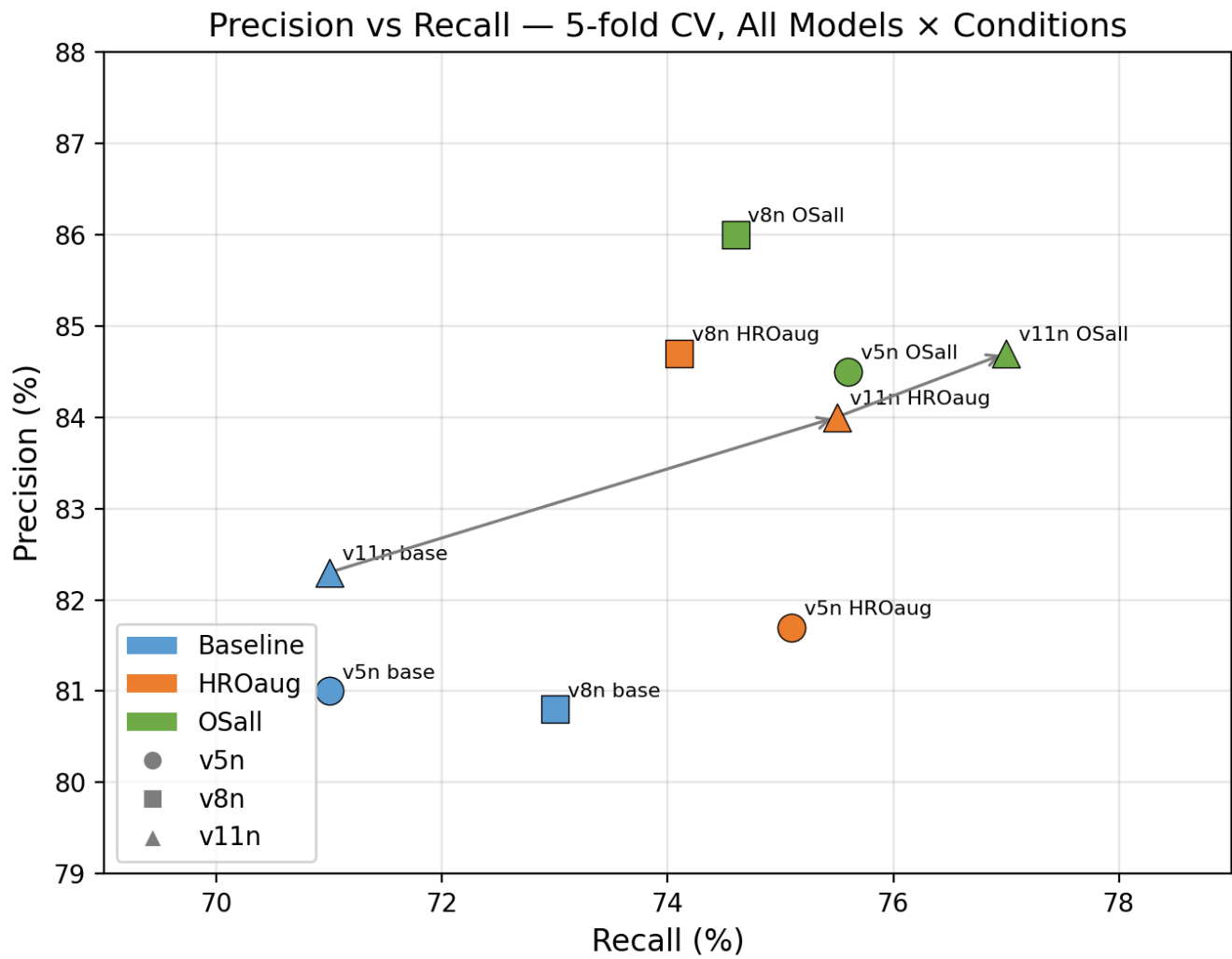


Fig 4: Precision vs recall. Arrows show v11n progression. All improvements move upper-right (better P and R).

Annotation Tightening A/B: Insulator Classes (OSall v11, matched seeds)

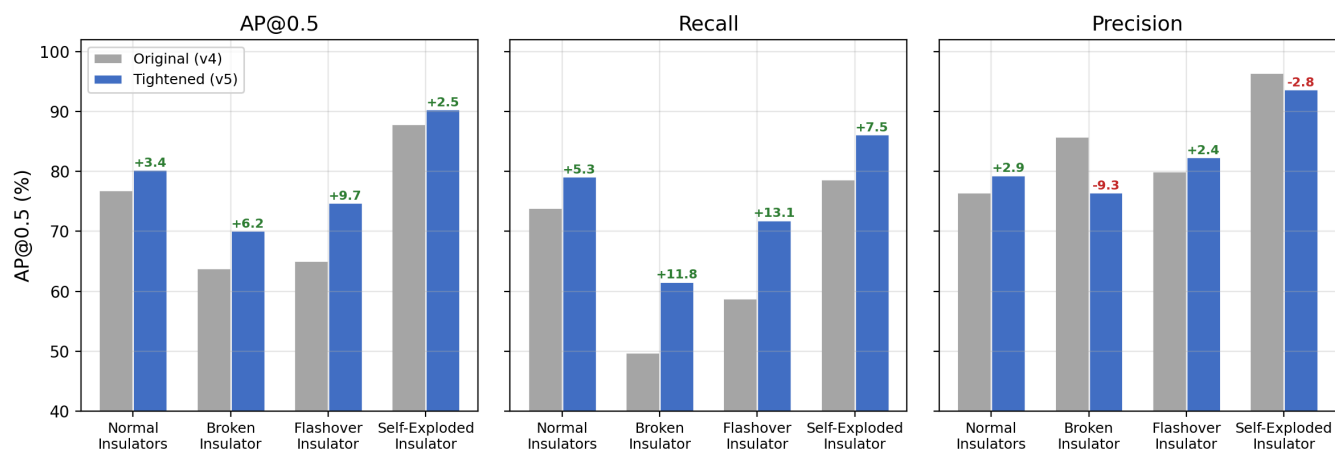


Fig 5: Impact of tightening Normal_Insulator annotations on insulator class recall.

Progression of Improvements — YOLOv11n

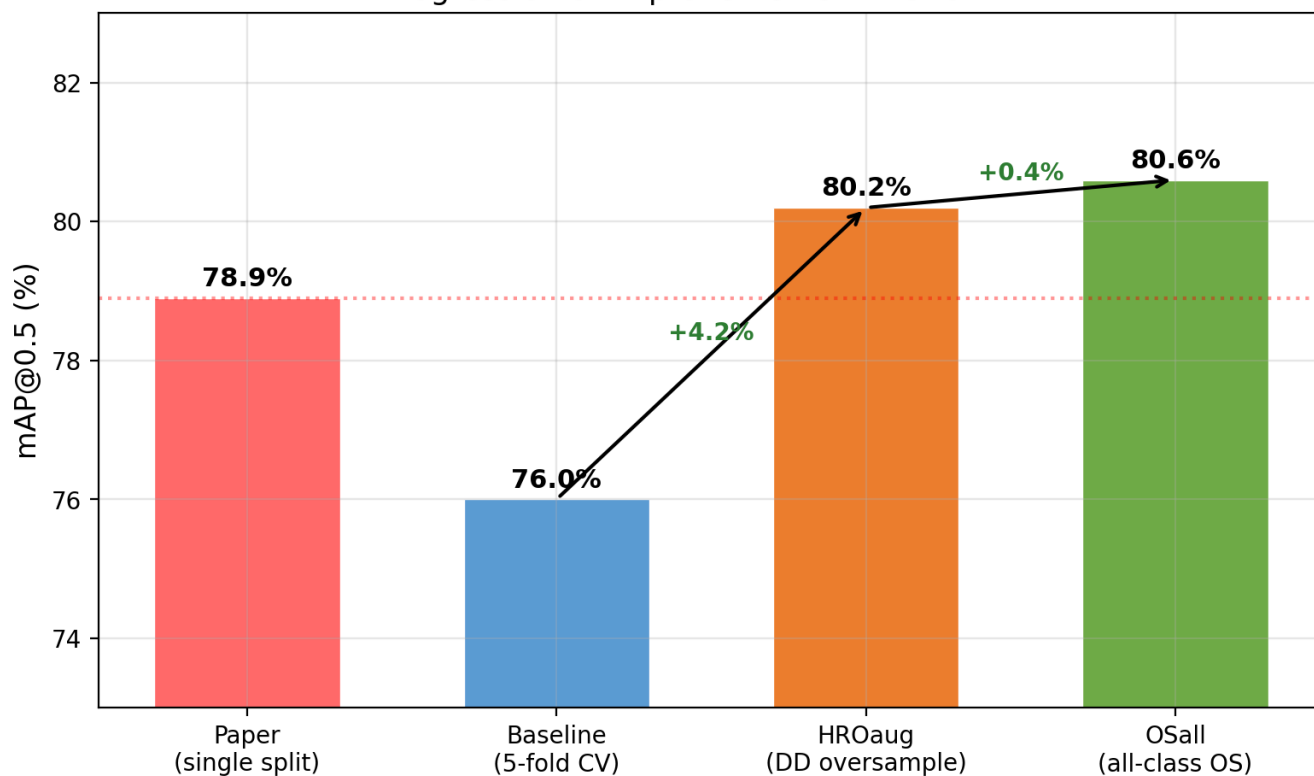


Fig 6: mAP progression from paper baseline to champion recipe.

1. Cross-Model Benchmark

Ran the full comparison across YOLOv5n, YOLOv8n, and YOLOv11n with three recipes:

- **Baseline:** same as the paper (640px, COCO transfer learning, 150+100 epochs)
- **HROaug:** 1280px + 3x oversampling of Defective_Damper + scale-down augmentation (scale=0.9)
- **OSall:** same as HROaug but oversampling all four minority classes 3x each

5-Fold CV Results

Model	Condition	mAP@0.5	Precision	Recall	DD AP	BI AP	FI AP	ND AP
v5n	Baseline	75.5%	81.0%	71.0%	72.1%	64.2%	61.4%	74.2%
v5n	HROaug	79.3%	81.7%	75.1%	75.0%	68.8%	67.9%	78.1%
v5n	OSall	79.8%	84.5%	75.6%	76.2%	69.1%	70.7%	77.8%
v8n	Baseline	76.5%	80.8%	73.0%	71.3%	64.8%	65.2%	74.7%
v8n	HROaug	79.9%	84.7%	74.1%	76.2%	69.0%	69.7%	78.0%
v8n	OSall	79.5%	86.0%	74.6%	75.3%	70.3%	70.0%	77.4%
v11n	Baseline	76.0%	82.3%	71.0%	72.7%	68.0%	61.4%	73.3%
v11n	HROaug	80.2%	84.0%	75.5%	74.5%	69.4%	71.9%	78.9%
v11n	OSall	80.6%	84.7%	77.0%	77.1%	71.8%	72.4%	78.1%

35 training runs total (3 models x 3 recipes x 5 folds).

Defect Class Detail: Precision / Recall / AP (5-Fold CV)

Model	Condition		Broken_Insulator			Defective_Damper	
		P	R	AP	P	R	AP

Model	Condition		Broken_Insulator			Defective_Damper	
v5n	Baseline	69.6	57.8	64.2	81.6	66.5	72.1
v5n	HROaug	72.8	60.8	68.8	83.0	72.1	75.0
v5n	OSall	81.6	61.4	69.1	82.5	76.0	76.2
v8n	Baseline	75.4	60.2	64.8	77.4	67.2	71.3
v8n	HROaug	74.2	64.3	69.0	85.1	69.6	76.2
v8n	OSall	91.7	61.1	70.3	84.0	72.6	75.3
v11n	Baseline	77.3	58.9	68.0	83.4	68.0	72.7
v11n	HROaug	80.5	60.0	69.4	81.4	72.8	74.5
v11n	OSall	86.2	63.0	71.8	85.5	73.2	77.1

Key findings:

- The improved recipe lifts every model by +3.4 to +4.6% mAP over baseline
- YOLOv11n is the best model across all conditions (80.6%), reversing the paper's finding that v5n was best (likely split-dependent)
- All three models with HROaug or OSall beat the paper's 78.9% in CV
- Defect class recall improvements (baseline to OSall, v11n):
 - Defective_Damper: 68.0% to **73.2%** (+5.2%)
 - Flashover_Insulator: 53.4% to **67.9%** (+14.5%)
 - Self-Exploded_Insulator: 78.6% to **87.6%** (+9.0%)
 - Broken_Insulator: 58.9% to **63.0%** (+4.1%)
- Precision stayed high or improved across all defect classes

2. Insulator Classes

Broken_Insulator (AP 50.6% at baseline on single-split) is the worst class in the dataset, worse than Defective_Damper (64.1%). It has only 194 instances and is visually similar to

Normal_Insulators.

OSall oversamples it alongside the other rare classes:

Class	Baseline	OSall	Improvement
Broken_Insulator	68.0%	71.8%	+3.8%
Defective_Damper	72.7%	77.1%	+4.4%
Flashover_Insulator	61.4%	72.4%	+11.0%
Self-Exploded_Ins	85.5%	92.4%	+6.9%

Flashover_Insulator had the largest per-class gain at +11.0%.

3. Annotation Refinement: Tightening Normal_Insulator Boxes

Tightened bounding box annotations for Normal_Insulators in the Roboflow target dataset (v4 to v5). A/B comparison uses OSall v11 with matched seeds (10, 11).

Overall (2-seed mean)

Annotations	mAP	Precision	Recall
Old (v4)	76.4%	85.4%	69.2%
Tightened (v5)	76.0%	81.6%	71.6%

Per-Class P / R / AP (2-seed mean, OSall v11)

Class		Old (v4)			Tightened (v5)		Delta AP
	P	R	AP	P	R	AP	

Class		Old (v4)			Tightened (v5)		Delta AP
Normal_Insulators	76.5	73.9	76.9	79.4	79.2	80.3	+3.4
Broken_Insulator	85.8	49.8	63.9	76.5	61.6	70.1	+6.2
Flashover_Insulator	80.0	58.8	65.1	82.4	71.9	74.8	+9.7
Self-Exploded_Ins	96.5	78.7	87.9	93.7	86.2	90.4	+2.5
Defective_Damper	89.7	67.4	73.6	76.1	51.1	58.9	-14.7*
Normal_Damper	85.8	77.2	84.1	85.7	73.1	79.4	-4.7*
Birdnest	83.6	78.8	83.6	77.5	78.3	78.3	-5.3*

* these classes are tested on different images. The insulator gains are the meaningful signal since those annotations actually changed.

Key finding: Tightening normal insulator annotations also improved defective insulator classes. Broken_Insulator recall went from 49.8% to 61.6% (+11.8%) and Flashover recall from 58.8% to 71.9% (+13.1%). Cleaner normal annotations give the model a sharper decision boundary for distinguishing defective variants.

4. External Data: Formally Ruled Out

2,175 vetted universe damper images were tested at every dose and configuration. None improved Defective_Damper. They consistently caused a recall drop.

Root cause: label/domain mismatch. The community "defective damper" is a different visual concept (different zoom, different defect definitions). An in-domain control (testing universe data on itself) scored 91% DD AP, proving the labels are learnable but do not transfer.

Negative result, but helpful.

5. Comparison to the Paper

Setup	mAP@0.5	Notes
Paper (v5n, single split)	78.9%	No CV, one split
Our baseline (v11n, 5-fold CV)	76.0%	Same recipe, robust evaluation
Our OSall (v11n, 5-fold CV)	80.6%	+4.6% over baseline

The paper's 78.9% was on a single split. Our baseline scores 76.0% in CV, suggesting the paper's split was favorable. Our improved recipe beats both numbers.

6. Next Steps

- **CV on tightened annotations** to get clean numbers for the annotation improvement on all classes
- **Change oversampling** Try 5x oversampling on broken insulator.